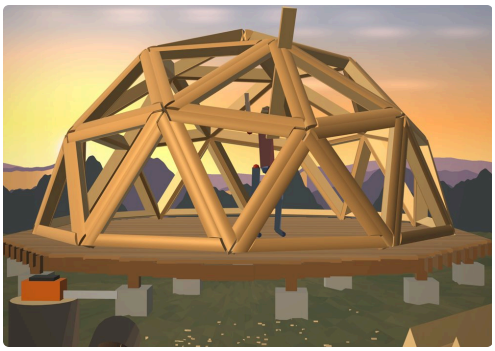
ALTERNATIVE HOUSING · OPEN SOURCE · OWNER-BUILT

The 40 Hour Cabin

Shelter you can make from the trees already standing.

One small chainsaw and about a working week to frame it: 120 split-log members, 40 flat panels from one jig, 277 square feet of floor — and every figure on this page computed by open-source software anyone can run.



37.5 h
to frame it: splitting measured, the rest estimated, 7.5 h kept for mistakes
PART MEASURED

$43.45
a sq ft for the standard building, margin in. Published: new house $217, manufactured $87
MODELLED

88%
of a log becomes frame when split; 45% sawn into two-by-fours
SIMULATED

42%
less outside skin than a box house of the same floor
MODELLED

15
buildings from one body, $12,036 to $25,511
MODELLED

15–22 kg
each strut, weighed: one person carries it
MEASURED

THE PROBLEM

Shelter eats the time and money a family needs for everything else: a new site-built house costs **$217 a square foot** before the land, a new manufactured home **$87** (NAHB, Census). And a tree passes through mills, trucks, yards and shelves before one board of it reaches a house.

THE IDEA

Skip the chain. Fell a standing tree, or take one already down, and rip it where it lies with **one small chainsaw**: each log section into 8 wedges, every cut through the centre, no angle measured. A two-frequency geodesic shape needs only two strut lengths, so 120 members become 40 flat triangles from one jig, raised into a 19.4-foot cabin. The triangles hold the shape; rough wood is good enough.

WHY IT WORKS

- **Rough wood is enough.** A triangle cannot change shape unless a stick changes length, so the frame forgives a wedge's rough faces. Storm-fallen and low-grade trees qualify.
- **Two lengths, one jig.** Every panel is built flat at waist height on the same jig, not up a ladder; the accuracy lives in the jig, not in the builder.
- **Money.** The frame is $692 of the bill; the standard building lists at $12,036, its margin printed on the invoice.
- **Shape.** Heating and cooling modelled at $129 a year, against $220 for a box of the same floor.

ONE BODY, MANY BUILDINGS

Stem cell · Gym · Studio · Nursery · Guest house · Workshop · Garage · Home · Sauna · Food · Advertiser · Cold storage · Bunker · Treehouse · Jacuzzi

What goes inside makes it a home or a workshop. It stands on a serviced pad and can move: **own the building, rent the ground** — with a network of pad hosts, and quilters paid to sew its insulation from recycled clothing.

WHO

Donovan Zeanah. U.S. Navy veteran: six years as a Mass Communication Specialist, three of them in Japan. Learned to program and was certified as an avionics bench technician; now studying for an A&P certificate on the G.I. Bill, on the way to aerospace engineering. Built so far, in the open: the solver, two books (83 and 55 chapters), 34 films and a family edition.

WHY I AM DOING THIS

> *Buckminster Fuller wanted housing cheap enough that a mother is not so burdened by it she cannot raise her child — what I took from him, not his words. I grew up with a single parent and a repossessed trailer, so I design to one question: what would a small home I owned have done for my eighteen-year-old self?*

A dome is a home written like good code: modular, reusable, decoupled, meant to be extended. **Every number is published with the code that computes it — including the ones that argue against it.**

BUILT TO BE LEARNED AT HOME

The method is taught in 6 rungs — from a bedtime story a parent reads aloud, through paper models, a plan and the build, to a family's own design and a published experiment — with 14 milestones to tick off together.

MEASURED, MODELLED, NOT YET

MEASURED	splitting, 30 wedges per 4-hour session; strut weights; a full-size prototype frame.
MODELLED	cost, energy and wood recovery, in open code anyone can run.
NOT YET	an engineer's rating for wind and snow; an instrumented year of weather. The first $12,850 below buys both.
AGAINST US	53% of the floor has 6.67 ft of headroom; the whole building is about 67 hours, not forty.

THE ASK: $100,650

The sum of its lines, not a round number: prove it, then make it repeatable.

Prove it — A full test platform and one standing dome · Sensors in the wall · An engineer	**$12,850**
Make it repeatable — CNC router · CNC plasma table · Laser cutter · Tooling to mould a strut with its hardware in it · Automated processing for the repetitive work	**$60,800**
Build what an owner cannot — Utility hub · Composite triangle	**$22,000**
Seed the network — Seeding the quilt network	**$5,000**
Total	**$100,650**

Costed as the next step, not asked here: a first site of 3 serviced pads, $100,000.

Every figure computed 13 October 2026 by the project's own software and labelled by kind.
Code, books and films: **github.com/dkzeanah/DomeSimulator**

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